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SOUTH AUSTRALIA.

THE IRON ORE RESOURCES OF THE MIDDLEBACK RANGE
SOUTH AUSTRALIA.

A SUMMARY

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INTRODUCTION

The Middleback Range lies approximately 26 miles west of the shipping port of Whyalla on Spencer Gulf, South Australia and forms a broken line of prominent ridges standing above a low plain. The hills are distributed meridionally and extend discontinuously with broad erosion gaps from Iron Knob in the north to Iron Duke in the south, a total distance of nearly 40 miles. They are composed primarily of iron rich rocks in which occur local concentrations of high grade iron ore.

At the present time one of these deposits, the Iron Monarch, the only iron deposit being worked in South Australia, supplies over 95% of the iron ore being utilised for iron and steel production in Australia. Present pre production from this deposit is of the order of 2,500,000 long tons per annum.

All known high grade iron ore deposits in the Middleback Range are controlled by the one company, the Broken Hill Proprietary Company Limited, which operates a blast furnace for production of pig iron at Whyalla, South Australia, and through its subsidiary companies controls most of the Australian iron and steel production, at Newcastle and Port Kembla in New South Wales. The Middleback Range iron ore deposits were first used solely as a flux in copper, and later lead smelting. They were taken over by the Broken Hill Proprietary Company in 1897. Up to the end of the last century it is estimated that about 100,000 tons of iron ore was mined for flux. From then on to the end of 1914 approximately 750,000 tons of iron ore was used for flux in lead smelting operations at Port Pirie, South Australia. Since the beginning of 1915, when the first steel

works was established at Newcastle, New South Wales, about 45,250,000 tons of iron ore has been produced from four deposits in the Middleback Range, viz., Iron Monarch, Iron Knob, Iron Prince and Iron Baron. At the present time ore quarried at the Iron Monarch is carried to Whyalla in rail trucks and thence loaded into ships for transport interstate to Newcastle and Port Kembla.

During the period September 1948, to December 1951, a departmental geological survey of about 2160 square miles of country in the Middleback Range area was carried out by the writer and his assistants. This included regional mapping on air photo scale of 35 chains to an inch of the whole area, detailed structural mapping of the entire Middleback Range on air photo scale of 900 feet to an inch and detailed plane table topographic surveys of all the scattered mining leases (covering about 3,800 acres) on either 100 or 200 feet to an inch. Object of the survey was to determine the distribution of the iron ore deposits in relation to the geology, hence if possible to decide on the probable method of formation of the orebodies so that estimations of the ore reserves in known deposits could be made, and the further search for new ore bodies planned. The following is a brief summary of the results of this survey.

PHYSIOGRAPHY

The Middleback Range area is one of generally mature, subdued topography and senescent drainage. The general plain level rises by a series of small and geologically youthful fault scarps to a height of about 400 - 450 feet above sea level at about 5 - 8 miles inland from the coast, and thence stands at about 450 - 650 feet for many miles to the westward. The Middleback Range comprises three major groups, of ridges, the most northerly being the Iron Knob - Katunga Hills group and the others the North and South Middlebacks, separated by Kimba Gap. Between the Iron Knob - Katunga Hills and the North Middlebacks, a distance of about 14 miles, is a

connecting line of low narrow ridges which arch to the westward and finally die out south of Camel Hill, about 1-1/2 miles west of Iron Baron in the North Middlebacks. This is known as the Camel Hill Line.

The highest points on the Middleback Range stand at 1450 - 1480 feet above sea level. Average height of the ridges is 1000 - 1100 feet and 350 - 550 feet above general plain level. The hills are usually bare and stony and flanked by rubble talus slopes. The plains are soil and rubble covered, and sparsely timbered, usually carrying a fair growth of low scrub, salt bush and blue bush and are used mostly for sheep grazing. Rock outcrops are scattered and rare. The climate is semi arid, average annual rainfall is 7-1/2 to 9-1/2 inches and annual evaporation 80 - 90 inches.

GEOLOGY

Stratigraphy

Several ages of geological formations or groups of formations have been distinguished as a result of the present survey of the Middleback Range area viz., Quaternary - Recent, Tertiary, ? Cambrian, Proterozoic and Archaean. The purposes of the present summary report will probably best be served by indicating the detailed stratigraphical sequence in the following tabulated form, together with a few additional explanatory notes. From then on the discussion of geological details will be confined to the economically important iron bearing rocks.

Table I Stratigraphic Sequence - Middleback Range Area

Quaternary - Recent	- Sand, soil, alluvium, dunes, travertine
Tertiary	- Laterite gravel, siliceous cement, grit and sandstone, marine limestone (Pliocene)

Unconformity

?Cambrian-	Corunna Group -	{	Lincoln Gap Flagstones - Flaggy, shaly sand-
			stones and interbedded shales.
			Treglana Shales - Laminated sandy and flaggy shales.

Table I Cont'd

Unconformity Cont'd

? Cambrian-	Corunna Group	{	Corunna Conglomerate - Conglomerate grit sandstone with some interbedded thin sandy shales.
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Unconformity

Proterozoic - Adelaide - Moonabie Grit. Dense felthspathic System
grit, coarse and fine sandstone quartzite.

Assumed unconformity

Archaean or middle Pre Camb- rian	{	Middleback Group	{	Upper Middleback Quartzite - Banded iron formation and interbedded schists.
				Cooks Gap Schist - Foliated grits and pebble schists phyllite with cherty and dolomite bands
				Lower Middleback Quartzite - Banded iron formations and interbedded schists.
				Middleback North Dolomite - Cherty dolomite marble.
				Gneiss Complex - Sedimentary schists, quartzite, phyllite, migmatite, lit par lit gneiss, granitic gneiss and gneissic granite.

Intrusive Igneous Rocks.

Quartz veins and reefs.

Younger Dolerite dykes.

Porphyritic Granite - Charleston Granite.

Gawler Range and Moonabie Porphyries.

Gneissic Granites and tor granites of the Gneiss Complex.

Older Dolerite - dolerite gabbro amphibolite - sills dykes and plugs.

The oldest rocks in the area apparently comprise a considerable sequence of conformable marine and shallow water sediments presumably laid down around the shores of a primeval land surface. These rocks were later subjected to extensive volcanic activity and penetrated by basic rocks (the Older

Dolerites) and then thrown into a series of tight anticlinal and synclinal folds before being uplifted and eroded. Towards the close of this period of folding the formations were fractured and penetrated by rising acid solutions and gases and granitic fluids which gradually replaced much of the sedimentary material from below, converting it to metamorphic schists, migmatites and lit par lit injection gneisses which have been grouped under the general name of Gneiss Complex. The overlying and unreplaced remnants of the originally extensive sedimentary series are known as the Middleback Group. It is the Middleback Group of rocks which constitute the iron bearing Middleback Range and Camel Hill line of ridges.

Apparently younger rocks believed to belong to the Adelaide System occur some distance to the east of the Middleback Range. These are steep dipping coarse grits and sandstones belonging to the Moonabie Grit formation which in turn is overlain with strong unconformity by gently tilted coarse to medium fine sediments of the Corunna Group, of probable basal Cambrian age. Pebbles in the Corunna conglomerate include sandstone - quartzite and grits of the Moonabie Grit and banded iron formation and iron ore of the Middleback Group.

The Moonabie and Gawler Range Porphyries are intrusive into the basal members of the Corunna Group, and the Charleston Granite is an even younger coarse, porphyritic granite boss. The Younger Dolerite is believed to be younger than all of the previously mentioned rocks.

Tertiary lateritic and siliceous gravels mark the remnants of a very extensive peneplain, the product of long continued erosion, and marine limestone marks. The position of a Pliocene marine incursion to the east of the Middleback Range. Quaternary - Recent soil and alluvium from senescent drainage lines mantle most of the land surface to the east, north-east and north-west of Range while to south-east, south, south-west and immediate west of the Range the country is covered with long sand dunes trending northwest - southeast.

Middleback Group.

This group of formations contains all of the iron rich rocks of economic interest in the present survey. The lowest recognizable horizon showing any degree of areal continuity in distribution is a bed of medium-grained dolomitic marble whose original deposition under reasonably deep water conditions apparently ushered in the period of deposition of the most important rocks, the banded iron formations. Two horizons of these banded iron formations, the Upper and Lower Middleback Quartzites, have been recognised in the South Middlebacks, in parts of the Camel Hill Line and the Katunga Hills (see Fig. 1). These two horizons are separated by a zone of original pebbly beds, grits and aluminous sediments now metamorphosed to foliated pebble schists and phyllitic schists (Cooks Gap Schists) having a thickness of up to 1200 feet.

The banded iron formations consist essentially of alternating thin silica-rich bands or strata in the form of dense chert, sometimes recrystallised to fine grained quartzite, and iron-rich layers in the form of predominating magnetite hematite, or martite. They are lithologically identical with the Pre-Cambrian banded iron formations of many other countries - the "taconites" and "jaspilites" of the Lake Superior region, the "itabirites" of Brazil, the "calico rocks" of South Africa, and the banded hematite quartzites of India, Sweden, Korea, Manchuria, North China, the Ukraine and the "jasper bars" of Western Australia.

In the Middleback Range the upper and lower horizons of banded iron formations, though thickened by plastic flow in folding are believed to have maximum thicknesses of up to about 1000 feet and 700 feet respectively. Due to folding and subsequent erosion the upper bed has been completely removed in the North Middlebacks. Both beds contain numerous interlayered thin shaly beds now metamorphosed to ferruginous chloritic or phyllitic schists. These banded iron formations are believed to be special forms of chemical sediments.

in which the iron and silica were transported as colloids and precipitated, under conditions of cyclical deposition influenced by a monsoonal type of climate, into closed marine basins or shallow lakes. The iron content varies from less than one percent up to fifty percent. Some of the bands are strongly magnetic and polarised. Many portions of the beds average about 30% Fe. and 55% SiO₂ and can be considered low grade siliceous iron ores (see below). Weathering of these rocks results in leaching of the iron in the form of limonite, leaving a siliceous gossan.

Geological Structure

Folding - The rocks of the Middleback Range have been thrown into a great number of synclinal and anticlinal folds whose axes trend slightly obliquely to the meridional direction of the Range itself. The major fold axis in both the North and South Middlebacks is synclinal with a major anticlinal axis passing down on the western side of the Range between it and the Camel Hill line of ridges. The axes of these major folds were themselves warped by later cross folding on a number of east-west axes probably whilst the beds were still in a slightly plastic state.

The result of this cross warping was to produce undulations in the pitch of the north-south fold axes, and to throw the formations into a number of alternating elongated dome and basin structures. Subsequent elevations and erosion of the whole area have resulted in the removal of most of the domed structures, leaving only the remnants of "keels" outlined by sections of the hard, highly resistant hematite-quartzite horizons. The Katunga Hills and North and South Middlebacks, are such remnants of compound keel structures, whilst the Kimba Gap and the area lying between the North Middlebacks and Katunga Hills, and the area south of the South Middlebacks represent regions which were once occupied by domes but being regions of tension were easily penetrated by granitic solutions and thus weakened have since been completely eroded away.

It is believed that this pattern of major folding together with its related minor fold structures had a very important influence on the formation and distribution of the high grade iron ore bodies of the Middleback Range. (Figs.1,2 and 3).

There is no evidence of the original distribution of or of folding in the Middleback Group of rocks beyond Ash Range to the east of the Middleback Range. It is possible that further search in the areas west of the Camel Hill Line may yet reveal evidence of the extension of folded structures of the formations in this direction. Such search can best be carried out in the first instance by aerial magnetometer surveys.

Faulting - Local faulting and shearing has been responsible for displacement of outcrops of the rocks of the Middleback Group in places but it is not considered that faulting played a major part in the formation of the Middleback Range as it is today, nor is it considered that shearing and faulting is a necessary precursor to the formation of the iron ore bodies. These views are opposed to suggestions made by L.K. Ward (1949).

THE IRON ORE DEPOSITS

Description

The principal high grade iron ore bodies of the area are the Iron Monarch and Iron Knob, Iron Prince, Iron Baron and Iron Duke, and there has been some production from all but the last. The Iron Knight, Iron Duchess, Iron Chieftain, Iron Queen, Iron Warrior and Iron Count are other smaller ore bodies in the area. Their distribution is indicated in the plan (Fig. 1).

Iron Knob - This deposit was the first to be developed and it has been quarried for a length of 1,600 feet and has a maximum width of 220 feet. This ore body is long and narrow, strikes north and south, and forms the core of a ridge. It apparently occupies the keel of a steep sided synclinal fold bounded by banded iron formation and altered ferruginous chloritic schists, and flanked to the east by granitic schists. The ore is uniformly high grade, usually fine grained massive hematite or martite.

It is typically low in manganese, phosphorus, silica and sulphur. Average partial analyses indicate 68.3% Fe, 0.07% Mn, 0.03% P and 0.05% S..

Iron Monarch - The Iron Monarch ore body is the present principal source of supply for the Australian iron and steel industry. The ore body occupies what was originally one of the highest hills in the region. It is roughly circular in plan and has a maximum north south length of about 2,100 feet and a maximum east west dimension of 2,200 feet. It has been developed by a series of benches for a vertical depth of approximately 350 feet below the original hill crest.

The ore body occurs in a broad open syncline modified by a central anticlinal ridge whose axes all trend slightly east of north and pitch gently to the north. The structure is now fairly well disclosed by a series of exposures on the various benches and roadway cutting as well as by some exploratory tunnelling and boring. The ore is flanked on the north, east and south by remnants of siliceous banded iron formations or hematite cherts and clay schists, and in places shows banded structures similar to that of the adjacent ferruginous cherts which it has evidently replaced. The transition zone is usually fairly narrow. The ore is cut by a series of three Younger Dolerite dykes striking north-west - south-east, the largest and most northerly of which forms the north wall of the ore in the present quarries. The general form of the Iron Monarch ore body and its irregular pitching floor is indicated by the geological sections =(Fig. 4).

The ore is high grade hematite with some residual magnetite or martite. It can be quarried to a grade of about 65% Fe. Manganiferous ore containing from 3% up to nearly 15% Mn originally formed a hard resistant core outcropping at the crest of Iron Monarch hill but much of this has now been quarried away. Clay-schist constitutes about 10% by weight of the ore body. Normally silica, alumina and phosphorus content is very low, and

titanium and sulphur are negligible. Average grade of high grade ore exposed in underground tunnels at Iron Monarch is about 64% Fe and 0.58% Mn. In addition there is a zone of mangiferous ore averaging 57% Fe and 8.6% Mn.

Iron Prince - This differs from the Iron Monarch in that the ore occurs not in a single mass but as a series of lenticular bodies intimately interfingering with the enclosing host rocks. These host rocks consist of banded iron formations and interbedded clay schists and abundant intrusive sills and dykes of dolerite-amphibolite all folded into a tight meridional synclinal whose keel pitches southerly at about 15°. The ore occupies parts of the central trough of this synclinal which is located in a composite ridge standing at a maximum height of 1100 feet or 400 - 450 feet above the surrounding plain. The keel of this trough is about 200 feet above plain level at the northern end of the Iron Prince but at the quarry in the southern end is probably more than 100 feet below plain level (See Fig. 5).

The ore has apparently been formed by replacement of the banded iron formations and schists, residual banding in the iron ore being identical with banding in the adjacent host rocks. The passage of replacing solutions has been controlled by the pre-existing synclinal structures in the banded iron formations. The ore is characteristically high grade, and low in manganese and phosphorus. Average composition of all ore so far produced is 64.7% Fe, 0.27% Mn and 0.028% P.

Iron Baron - This is a large prominent hill reaching a maximum height of 1250 feet above sea level. Much of the hill crest is covered with recemented limonitic iron ore rubble or "float". The hill carries a number of lenticular masses of dense banded hematite, the most important of which lies chiefly on the western slope of the hill where it occupies the keel of a synclinal fold in banded hematite quartzite and schist host rock.

So far all production from the Iron Baron has come from a small quarry at the northern end of the hill, the principal ore body not yet having been opened up. Within this quarry

on the western wall of the ore body there is a rapid transition from massive banded hematite ore to banded ferruginous chert or banded iron formation. Average grade of ore yet produced is 62% Fe, 0.22% Mn and 0.038% P.

Iron Duke - This ore body appears to be confined chiefly to the flatter dipping western limb of a north pitching syncline which is itself a large drag fold on the major synclinorium forming the South Middlebacks. The steeper, nearly vertical eastern limb consists of normal siliceous banded iron formations and the transition zone from high grade iron ore to banded ferruginous quartzite is quite narrow. Although this ore body has not yet been developed, from sampling it would appear to be similar in grade and mineral content to the other high grade ore bodies in the Range and should be capable of yielding ore containing 60-65% Fe, low in manganese and phosphorus.

Other Orebodies - Most of the other smaller bodies of iron ore in the Middleback Range are similar to the larger ore deposits already described, both in type, in composition and grade of ore, and in structural relationships to host rocks. With the possible exception of the Iron Knight, the iron ore in all of these smaller deposits appears to be located in pitching synclinal drag folds of larger or smaller scale, within the banded iron formations and associated argillaceous and ferruginous schists. As far as can be ascertained from mapping surface features, the ore lenses within the Iron Knight leases seem to occupy portions of a steadily dipping limb structure rather than a fold trough.

In general the banding of the high grade hematite ores is indicative of replacement of the banded hematite quartzite host. Corresponding replacement of the associated argillaceous schists or of the amphibolites and amphibolite schists has nowhere been so complete, these rocks usually tending to remain within the ore as "mullock" bodies. It is noticeable that the high grade iron ore deposits are confined to the banded iron formation horizons of the Middleback group and where the two horizons can be distinguished it has been found that the ore bodies occur within the

stratigraphically lower formation. No ore bodies have been discovered within the uniformly dipping limb structures of the Camel Hill line.

ORE GENESIS

Several conflicting theories as to the origin of the high grade iron ore bodies, in the Middleback Range have been advanced. These are:-

- (1) That they are lodes injected into the country rock from some external magmatic source.
- (2) That they have been formed by replacement of the country rock by iron introduced from some external source by the agency of ascending hydrothermal solutions.
- (3) That they have been formed by replacement of the country rock by iron leached from adjacent overlying^(outcropping) portions of the same country rock and transported by descending supergene solutions.

The first two theories, supported originally by R.L. Jack (1922) and L.K. Ward (1949) respectively fail adequately to explain much of the field evidence, particularly in regard to the common relationship of high grade ore bodies to certain rock types and rock structures and are thus considered invalid.

Theory No. 3 is the classical theory of downward leaching and migration under the action of meteoric waters which has been supported by many competent authorities to explain the origin of the major hematite ore bodies of the Lake Superior Region - such as the high grade iron ores associated with the "taconites" of the Mesabi Range. This theory explains satisfactorily many of the field facts regarding these ores - their apparent confinement to the troughs of pitching synclines within the banded iron formations, the proved absence of any roots or subterranean "feeders" entering the base of some of the ore bodies, the uniform chemical and mineralogical composition of the ore, and the general absence of sulphides and hydrothermal gangue minerals in the ore.

Features which are not satisfactorily explained by this theory are (a) the absence of any marked vertical variation in the quality or physical characters of the ore, (b) the absence of any ore bodies in the Upper banded iron formation horizon in the South Middlebacks where synclinal fold structures presumably suitable for ore concentration can be found, (c) the fact that under present day weathering conditions iron rather than silica is first leached, the iron being transported and deposited as limonite and the silica subsequently being removed by mechanical erosion processes.

A further hypothesis (No.4) is suggested by the writer viz., that the high grade iron ores were formed by replacement of the country rock by iron leached from adjacent portions of the same country rock, and concentrated by the agency of laterally moving hydrothermal solutions assisted by hydrothermally mobilised supergene waters. This theory depends on the view that hydrothermal activity associated with the formation of the Gneiss Complex by the granitic injection, metasomatism and feldspathisation of pre-existing sediments, took place after the deposition of the Middleback Group of sedimentary formations and was responsible for their metamorphism and conversion, in places, into granitic gneisses, migmatites and schists. The granitisation process is conceived as having been initiated along the axes of major anticlinal folds and gradually advanced laterally towards the folded synclinal zones. Leaching of the banded iron formations produced quartzites, and later feldspathisation converted them to gneisses. The leached iron advancing ahead of the front of granitisation and transported by hydrothermal and heated mobilized supergene waters, entered the remnant synclinal zones of Middleback Group rocks which now form the Middleback Range, and became concentrated within favourable structures to form the existing ore bodies. The stages of this process are illustrated diagrammatically in Fig. 6.

Final downward movement of these waters into pitching synclinal troughs would account for the shape of many of the ore

bodies and the apparent lack of channelways entering them from below. This hypothesis would also explain the apparent concentration of iron ore bodies into the Lower Middleback Quartzite horizon which, within the major keel structures tended to form a protective casing surrounding the Upper horizon. It would also explain the high degree of oxidation of the ore bodies, and why no repetition of the ore forming process has occurred in later geological time. The absence of sulphides is attributable to the strongly oxidising conditions of ore formation and the absence of normal hydrothermal gangue minerals to the highly dilute form of the circulating waters due to the abundance of meteoric solutions and to the extremely simple chemical composition of the replaced host rocks.

ORE RESERVES

Reserve Estimates

The following tabulated statements of ore reserves represent estimates based on the results of the recent field surveys. They have been subdivided into high grade iron ore such as has been described occurring in various mineral leases held by the Broken Hill Proprietary Coy. Ltd., and low grade ore comprising the siliceous banded iron formations of the Middleback Group encountered in the Middleback Range, the Katunga Hills, and the Camel Hill line. The estimates must be considered only tentative, based on individual judgment regarding the distribution of the ore in many places and will no doubt be subject to considerable revision when the field of underground exploration is extended.

The calculated reserves of high grade ore have been subdivided into "probable" ore and "possible" ore. "Probable" includes proved or "blocked out" ore together with the ore in those portions of the ore bodies which from geological evidence can be reasonably assumed to extend down at least to the limits of economic quarrying. "Possible" ore includes inferred or hypothetical extensions of the "probable" ore boundaries,

deduced from geological reasoning but of which direct evidence is inconclusive, together with those portions of ore bodies for whose presence there is adequate field evidence but which may be beyond the limits of economic quarrying. Much "possible" ore may probably be recovered by mining.

TABLE II Reserves of High Grade Iron Ore - Middleback Range

Ore Deposit	Probable(tons)	Possible(tons)	Total(tons)
Iron Monarch	59,245,000	13,131,000	59,376,000
Iron Knob	3,403,000	481,000	3,890,000
Iron Prince	16,362,000	6,396,000	22,758,000
Iron Baron	19,754,000	3,020,000	22,774,000
Iron Duke	17,510,000	16,919,000	34,429,000
Iron Duchess	4,340,000	7,404,000	11,744,000
Iron Knight	5,500,000	10,800,000	16,300,000
Iron Chieftain	1,200,000	1,300,000	2,500,000
Iron Queen	1,279,000	635,000	1,914,000
Iron Count	358,000	-	358,000
Iron Warrior	340,000	-	340,000
South Iron Queen	47,000	-	47,000
Totals	129,338,000	47,092,000	176,430,000

TABLE III Reserves of Low Grade Iron Ore - Middleback Range.

Deposit	Tons
Katunga Hills	388,000,000
North Middleback Range	2,906,000,000
South Middleback Range	1,879,000,000
Camel Hill line	257,000,000
Total	5,430,000,000

Grade

The general grade of the high grade iron ore bodies of the Middleback Range has already been indicated. The ores

vary little in composition in different parts of the range. They are typically almost free of sulphur, titanium and phosphorus, and seldom contain more than about 2% of combined silica and alumina. Manganese is the principal variant and may rise from less than 0.5% Mn to 15% Mn. There should be little difficulty in maintaining an ore grade of 64.5% combined iron and manganese from each of the high grade ore bodies cited above.

The low grade iron ore bodies, the siliceous banded iron formations, are probably more variable in proportions of iron to silica, but the minor constituents are characteristically extremely consistent. These features are indicated in the following assays of 4 representative samples from different parts of the Middleback Range:-

TABLE IV Composition of Low Grade Iron Ores - Middleback Range.

	A	B	C	D
H ₂ O	1.48	2.08	1.92	1.78
SiO ₂	51.24	51.78	53.04	56.52
Fe	33.10	32.15	31.11	28.92
Mn	0.07	0.07	0.08	0.07
P	0.05	0.05	0.04	0.05
S	0.02	0.03	0.01	0.04
Ti	Trace	0.003	0.003	0.003

A. Katunga Hills

B. Camel Hill

C. South Middlebacks - south side Kimba Gap

D. South Middlebacks - central range near Mt. Middleback.

FUTURE DEVELOPMENT

The future exploitation of the Middleback Range iron ore bodies is inevitably bound up in the future of the Australian steel industry. At the present time the steel making capacity of the country is well below the rate of steel

consumption and it is obvious that in a relatively undeveloped country with a steadily increasing population and a rapidly expanding industrial capacity, such as Australia, a steadily accelerating demand for steel can be anticipated.

S.B. Dickinson, in a recent review of the Australian steel industry, (1952) has estimated that by 1960 Australia's steel requirements will be at least 5,000,000 tons annually. This would require production of a minimum of 7,500,000 tons of iron ore annually or about 3 times the present rate of production. Dickinson has further suggested that a desirable expansion would be to an annual steel making capacity of 7,500,000 tons, requiring 11,250,000 tons of iron ore by 1960.

It can be anticipated that because of their purity and accessibility the Middleback Range iron ore deposits will continue to supply at least a half and more probably near threequarters, of the Australian iron ore requirements for some considerable time. Assuming a threefold expansion of ore production from the Middleback Range deposits during the next decade, very considerable inroads would have been made into the known reserves of high grade ore. Indeed, without the proving of further reserves, and assuming an expansion in ore production of the above order, by 1970 the effective future life of high grade ore in the Middleback Ranges could well have been reduced to no more than 15 years.

For continuation of steel production on the scale predicted beyond that time, if raw material from the Middleback Range area is to be utilised, it will be necessary to follow the lead given in America and other countries and develop the low grade iron ores (banded iron formations). For this purpose it will be essential to develop a satisfactory method of beneficiation of these low grade ores to convert them to a suitable grade and physical condition for feeding into the blast furnaces. No doubt the advances at present being made into the metallurgical techniques of iron ore beneficiation from the taconites of the

Lake Superior Region in North America will be closely watched by interested parties in this country.

Major problems are involved in the switch over from high grade to low grade iron ore utilisation however. For the same rate of steel production the ore production capacity must be increased threefold. Irrespective of what ultimate beneficiation process is evolved, milling plants must be established to treat the entire production of low grade ore and large capacity plants capable of up grading and agglomerating the milled products must be installed. This must entail an enormous capital expenditure. The total available reserves of low grade banded iron formations in South Australia and elsewhere (particularly Western Australia) are probably adequate to supply Australia's requirements irrespective of expansion, for many centuries to come, but the remoteness of many of these deposits from industrial centres will mitigate against their exploitation for a very long time.

It is apparent that the capital expenditure necessary for the transfer to low grade iron ore utilisation, however long delayed, must inevitably sharply raise the production cost of Australian steel and steel products. It would seem most advisable to cushion this cost rise by a very gradual change-over, spread over a very considerable number of years, so that the continuous parallel production of cheap high grade ore would for a long time be available to "sweeten" the market.

For this reason the writer is convinced that an early start should be made in planning for the establishment of the necessary beneficiation treatment plants and for the active mining development and exploitation of suitably located deposits of banded iron formation in Australia. He believes that within 10 - 15 years from now production and treatment of low grade iron ores on a working scale should have been fully established in this country, and should be making real contributions to the raw materials for the steel industry.

The low grade ores of the Middleback Range, being so

close to the routes for the transport of high grade ore, can be expected to receive early attention. The banded hematite quartzites from this region are in many cases chemically and mineralogically less complex than some of the taconites already being successfully processed in the Lake Superior Region, and should present no unique metallurgical problems in beneficiation.

CONCLUSIONS

As a result of the recent detailed survey of the geology and iron ore resources of the Middleback Range region the writer has reached the following conclusions:-

1. There are a limited number of very pure high grade iron ore bodies exposed in the Middleback Ranges. Most of these are reasonably accessible and capable of exploitation by surface quarrying methods. All known deposits of economic significance were found prior to the present survey and are being held for exploitation.
2. All known high grade iron ore deposits in the Middleback Range have been found associated with a type of low grade siliceous iron ore or banded iron formation similar to the "taconites" of the Lake Superior Pre-Cambrian iron ranges.
3. The high grade ore bodies have been formed by replacement of portions of the pre-existing folded banded iron formations and associated rocks. Incontrovertible evidence of the mechanics of this replacement has not been obtained, but it is suggested that the iron was derived by leaching from adjacent portions of the same banded iron formations and was concentrated through the agency of laterally moving mixed hydrothermal and heated supergene waters. The ores appear to have been deposited only in suitably disposed structural traps within the banded iron formations.

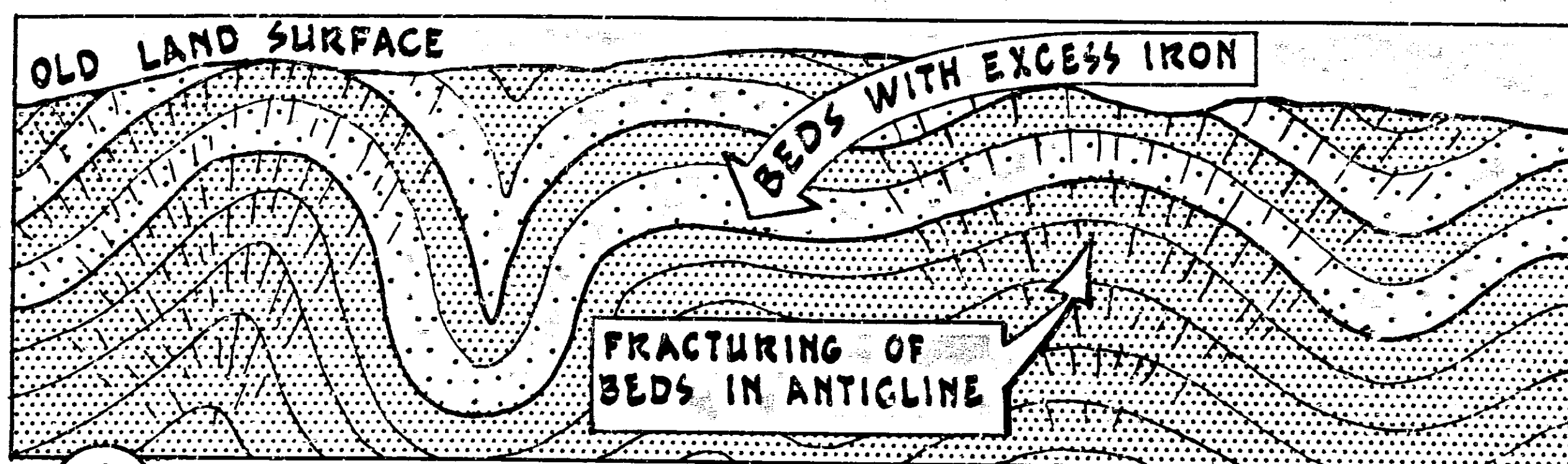
4. No new deposits of high grade ore have been discovered in the Middleback Range area during the course of the present survey.
5. The finding of any possible new high grade iron ore deposits of economic size will be contingent upon the discovery of new areas of banded iron formations. The highly magnetic character of these rocks make them excellent subjects for detection, on a regional basis, by means of the air borne magnetometer. There is a very large area of Eyre Peninsula potentially capable of containing rocks similar to the formations of the Middleback Group. For large scale exploratory work airborne magnetometer surveys should precede the more intensive prospecting by field parties.
6. The probable reserves of High grade iron ore within the Middleback Range are adequate for Australia's immediate requirements and at her present steel making capacity, but cannot be accepted as adequate as a long term source of raw material for a rapidly expanding steel industry.
7. Unless startlingly new discoveries of high grade iron ore are made in Australia during the next few years, early positive steps should be made to initiate the change over to low grade iron ore utilisation.
8. It is suggested that it would be within the national interest for the installation of a suitable beneficiation plant to have been completed, and production of iron ore concentrates from low grade banded iron formations to have begun, within the next 10 years.
9. Because of their uniform and simple composition and ready access to established centres of distribution the banded iron formations of the Middleback Range should have priority in plans for treatment and exploitation.

REFERENCES

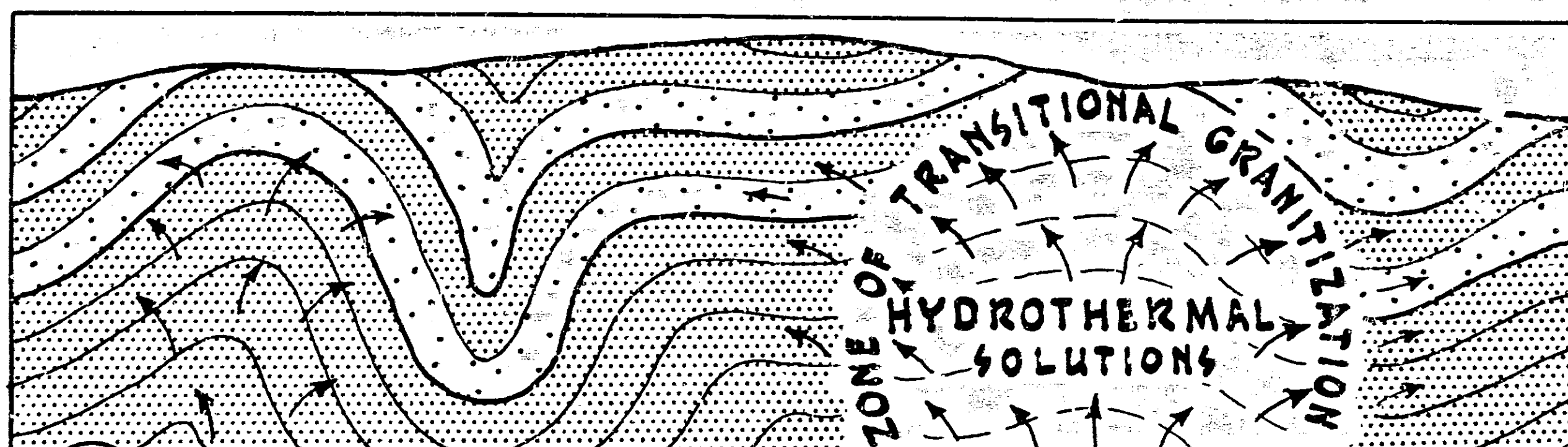
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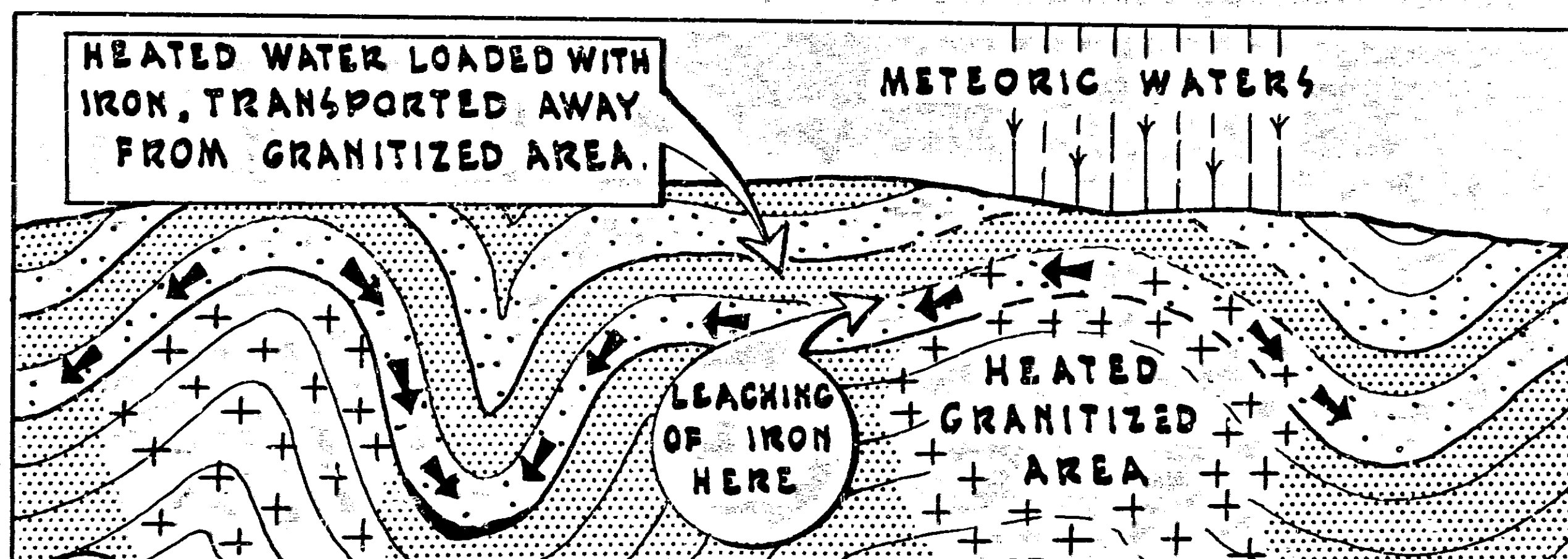
H. R. Miles
Senior Geologist.



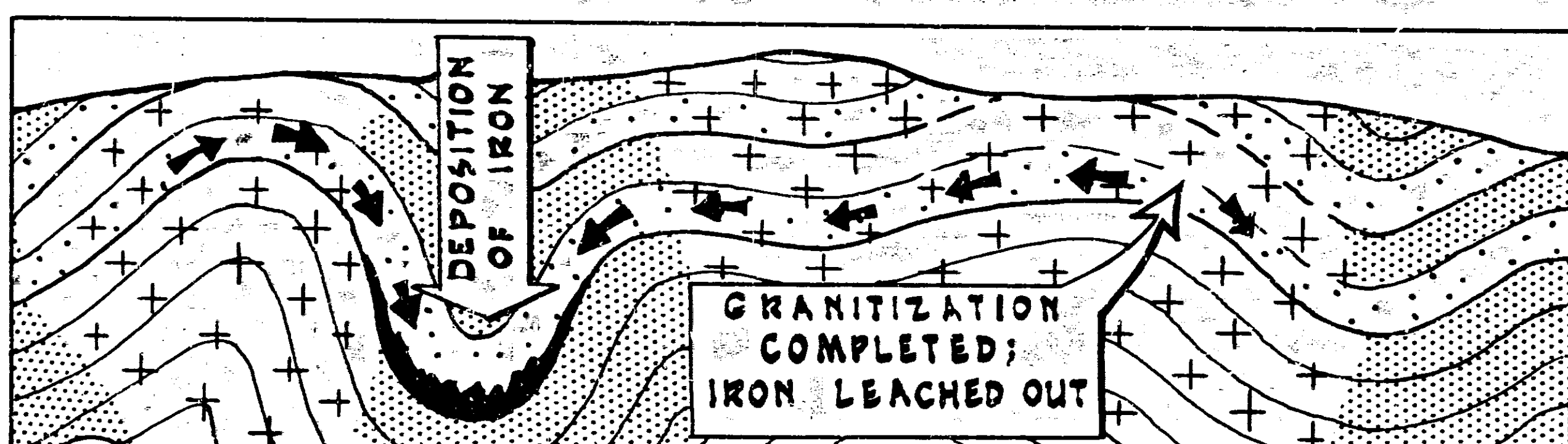
1 SHOWING OLD LAND SURFACE & ORIGINAL FOLDED STRUCTURE.



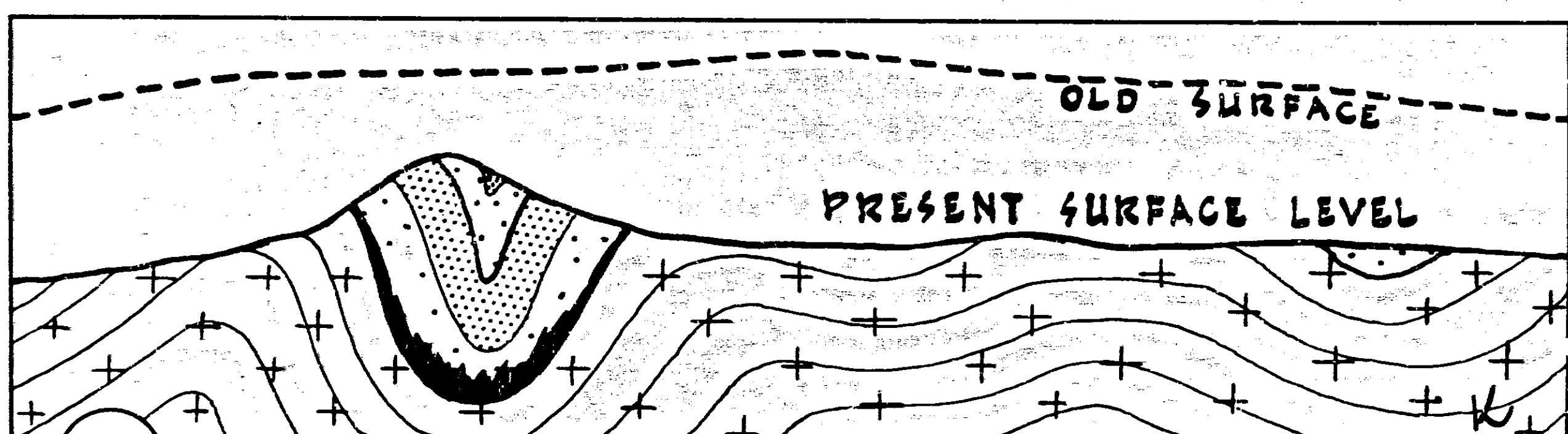
2 INTRODUCTION OF HYDROTHERMAL SOLUTIONS & GRANITIZATION



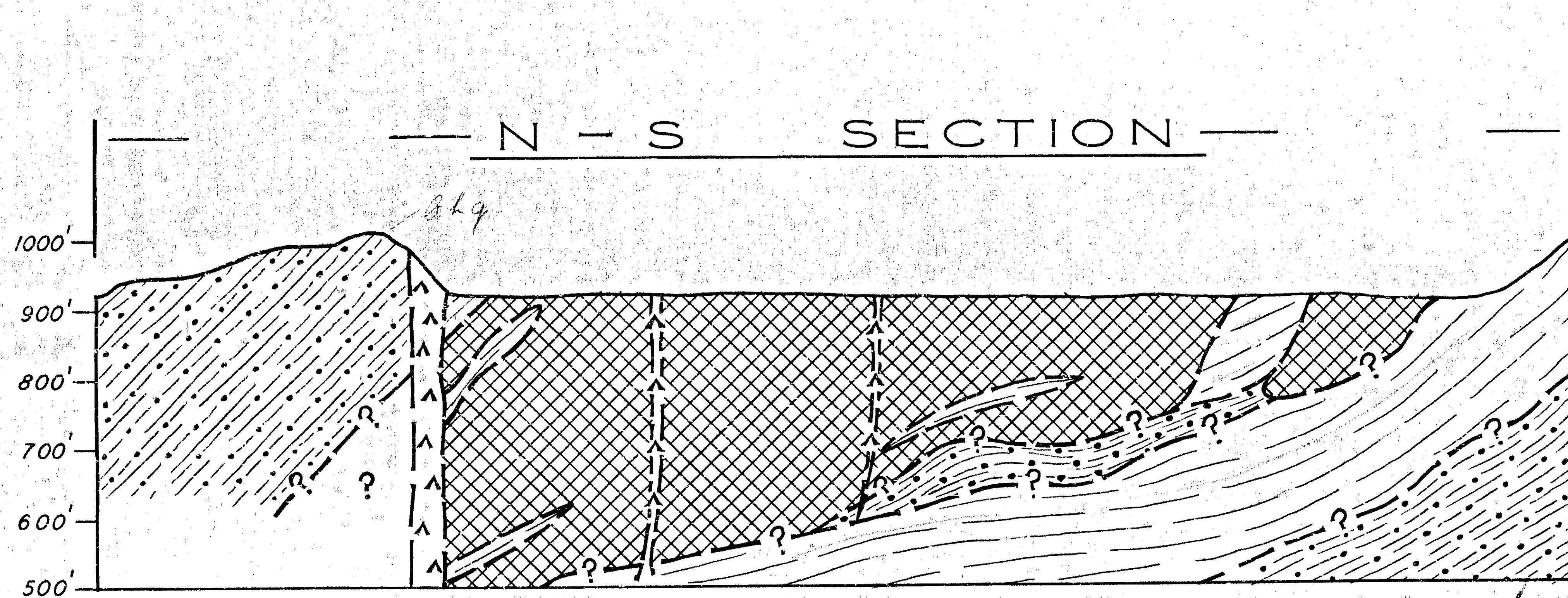
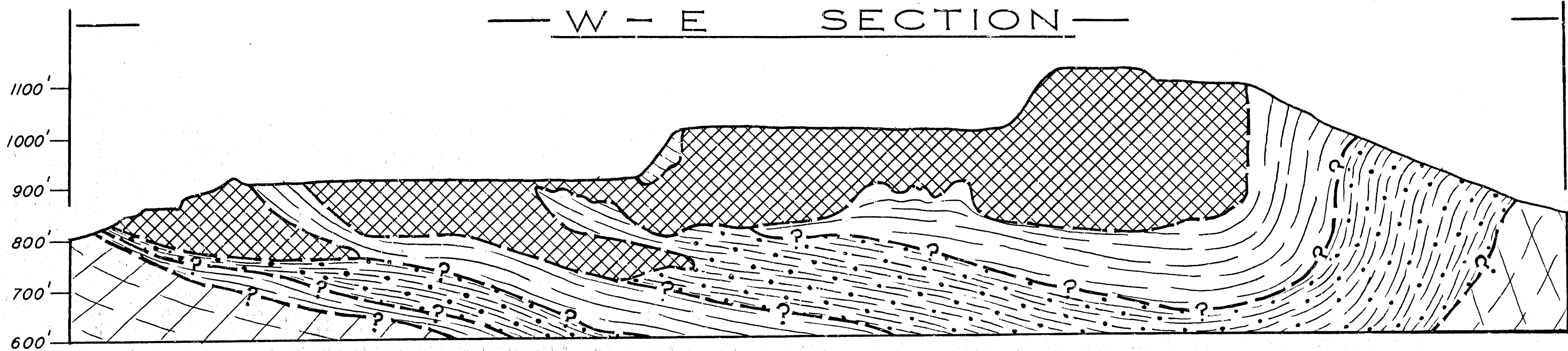
3 LEACHING AND MOVEMENT OF IRON BY HEATED WATERS.



4 IRON TRAPPED & DEPOSITED IN TIGHT FOLD OF SYNCLINE.



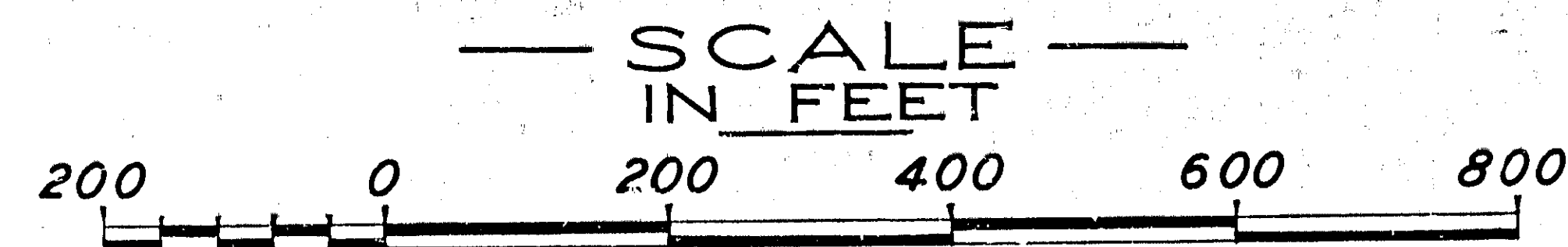
5 DEPOSIT IN RELATION TO PRESENT DAY LAND SURFACE.

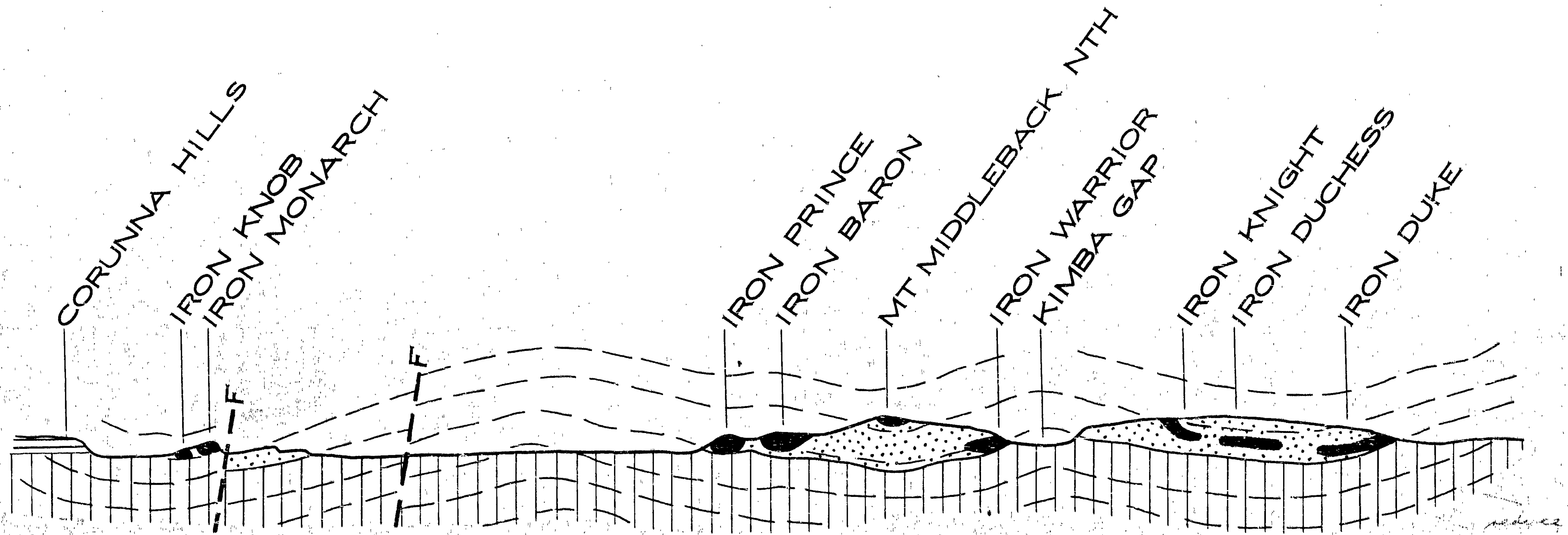


— LEGEND —

- High grade iron ore....
- Schists, with hematite bands.....
- Banded hematite quartzite with intercalated schist bands, and hematite...
- Granitic Schists.....
- Dolerite dykes.....

**IRON MONARCH ORE BODY
DIAGRAMMATIC SECTIONS**





LEGEND

Iron Ore



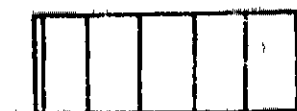
Middleback Group



Corunna Group

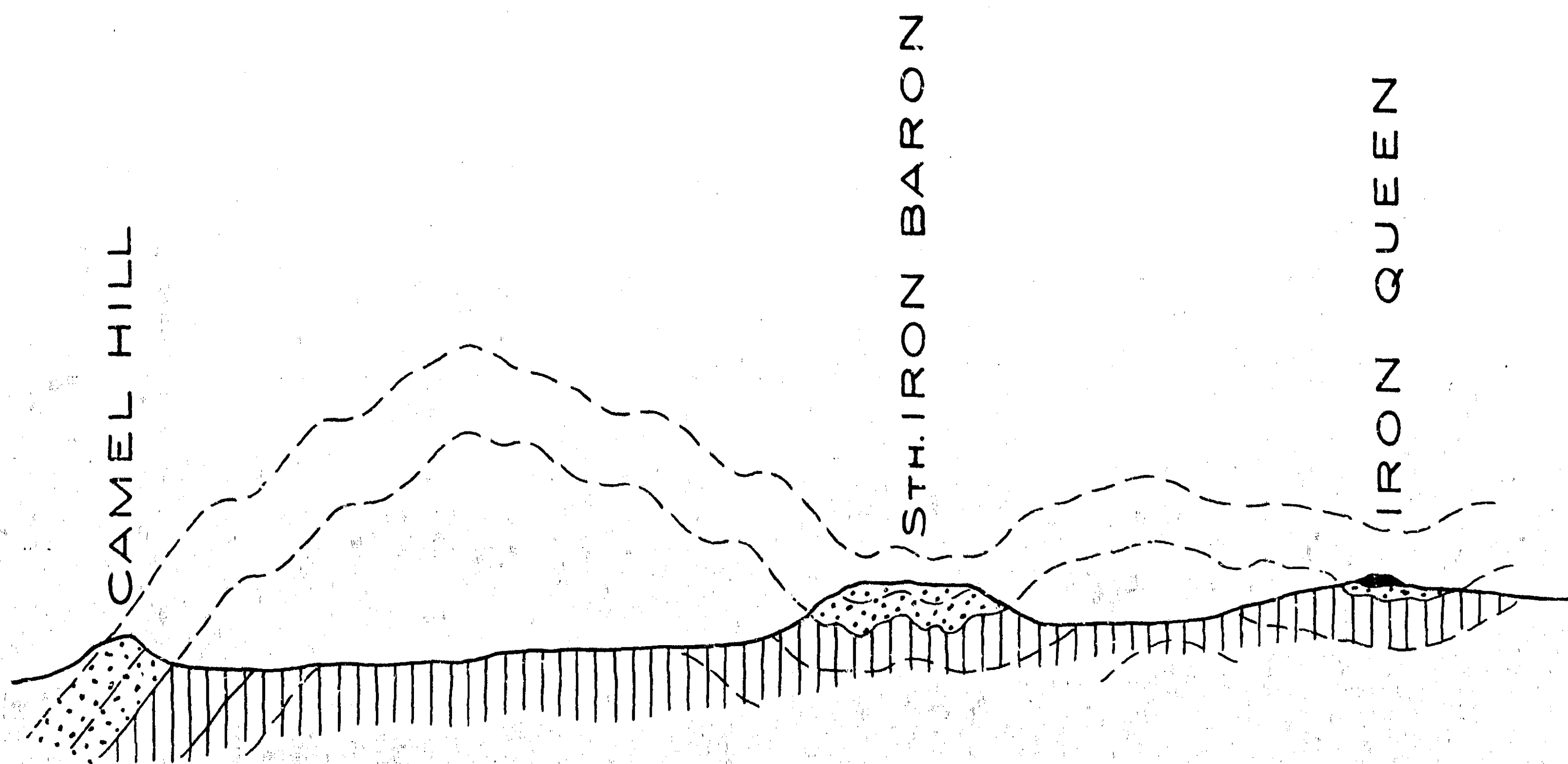


Gneiss Complex

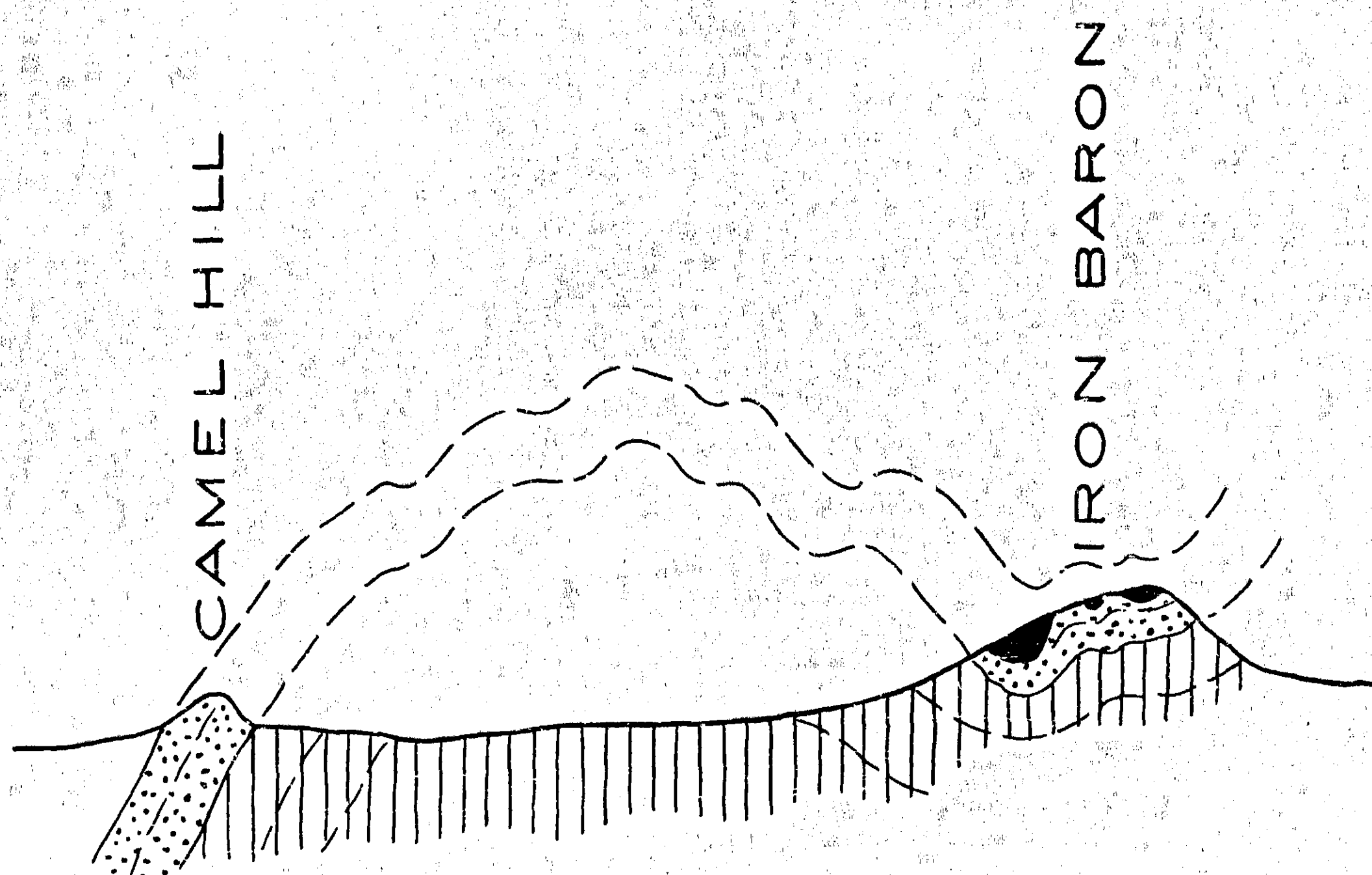


DIAGRAMMATIC LONGITUDINAL SECTION THROUGH IRON KNOB AND MIDDLEBACK RANGES

CAMEL HILL TO IRON QUEEN



CAMEL HILL TO IRON BARON

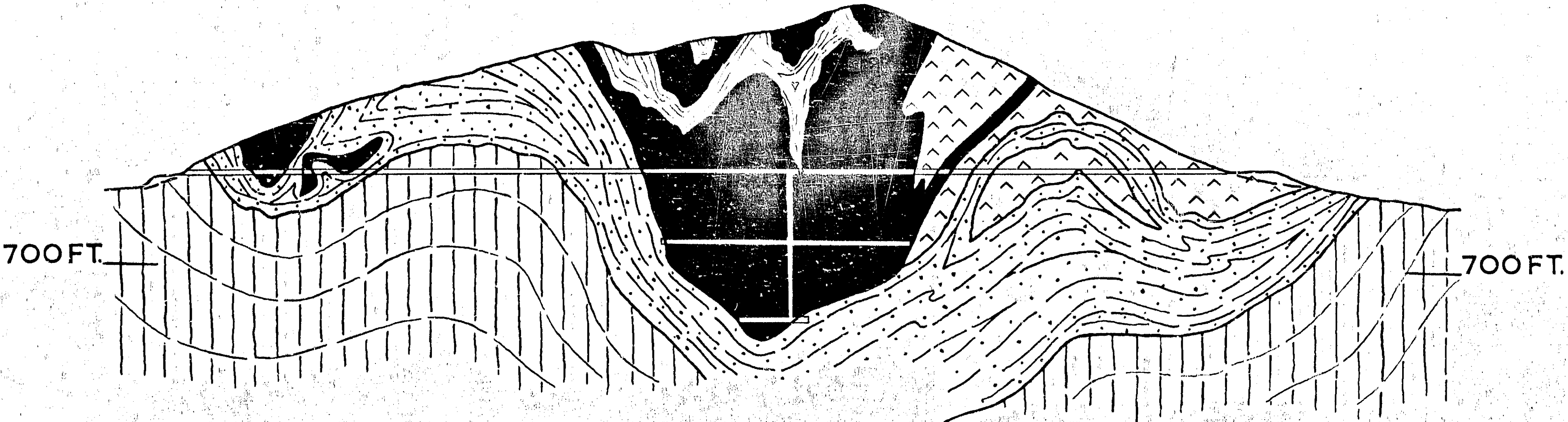


GEOLOGICAL SKETCH SECTIONS

LEGEND

IRON ORE		
MIDDLEBACK QUARTZITE		
GNEISS COMPLEX		

IRON PRINCE
GEOLOGICAL SKETCH SECTION THROUGH TUNNEL NO.1A



LEGEND

- IRON ORE _____
- RED SCHIST _____
- AMPHIBOLITE _____
- BANDED HEMATITE QUARTZITE _____
- GNEISS COMPLEX _____

